Dark Web

Agenda

- **Introduction**
- Review
- VPN proxies
- Tor
- Tor services
- Tor browser
- Tor history

The dark web

- Part of the Web accessible only through anonymous tools (e.g. Tor Browser)

SURFACE WEB

Google

Bing

Wikipedia

DEEP WEB

Academic Information

Medical Records

Legal Documents

Scientific Reports

Subscription Information

Contains 90% of the information on the Internet, but is not accessible by Surface Web crawlers.

Social Media

Multilingual Databases

Financial Records

Government Resources

Competitor Websites

Organization-specific Repositories

(DARK WEB)

A part of the Deep Web accessible only through certain browsers such as Tor designed to ensure anonymity. ***Deep Web Technologies has zero involvement with the Dark Web.***

Illegal Information

Drug Trafficking sites

TOR-Encrypted sites

Political Protests

Private Communications

Dark web vs. deep web

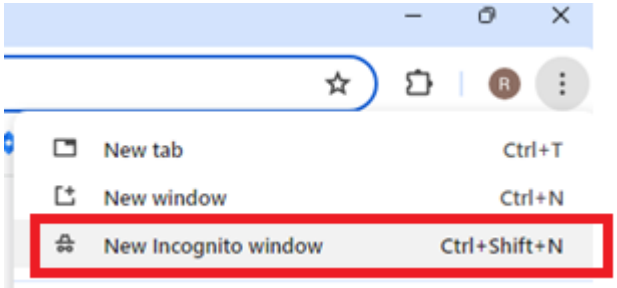
- *Deep web*: pages that search engines can't find
  - Pages behind paywalls, e.g., Wall Street Journal
  - Pages behind passwords, e.g., Gmail, Facebook, online banking

- *Dark web*: pages accessible only through anonymous tools

- The deep web includes the dark web

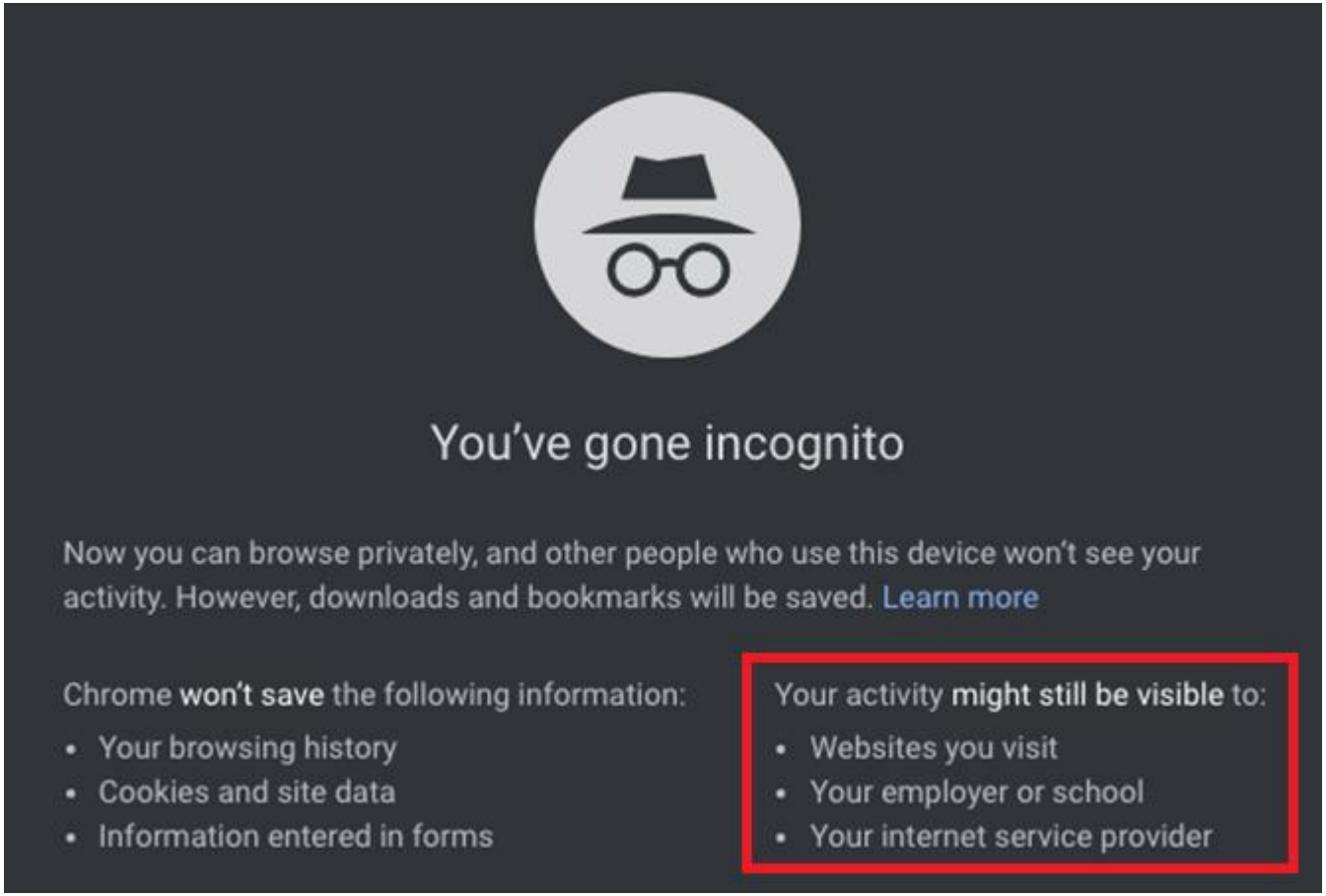
Anonymity online

- Is anonymous browsing mode enough for anonymity online?
  - Private browsing mode, incognito mode, etc.



- What about HTTPS?
- Is that enough? Why?

Anonymity online



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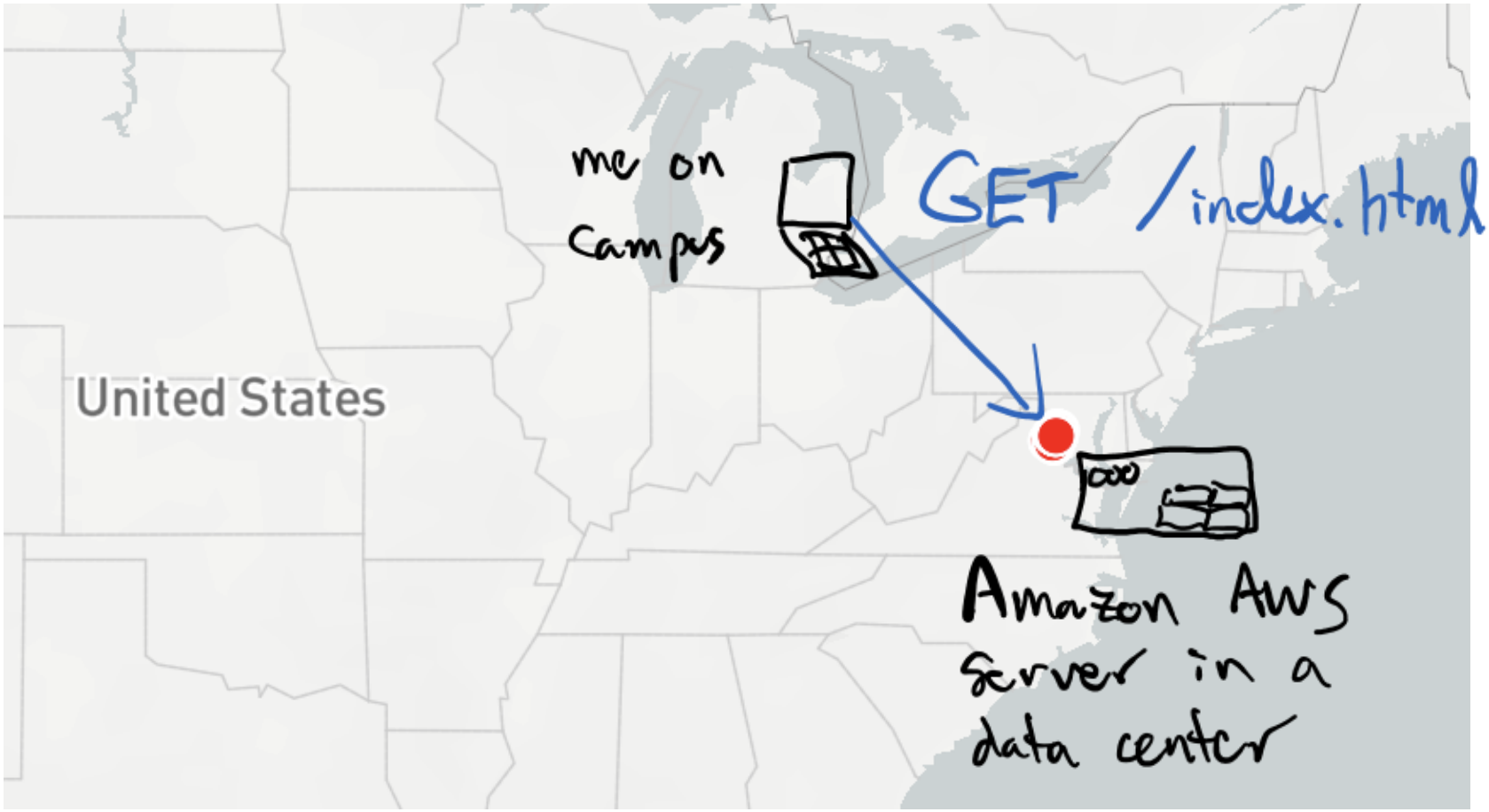
Review: Network communication

- Two parties communicate over the internet
- Example: client visits `amazon.com`

Review: Example message from client to server

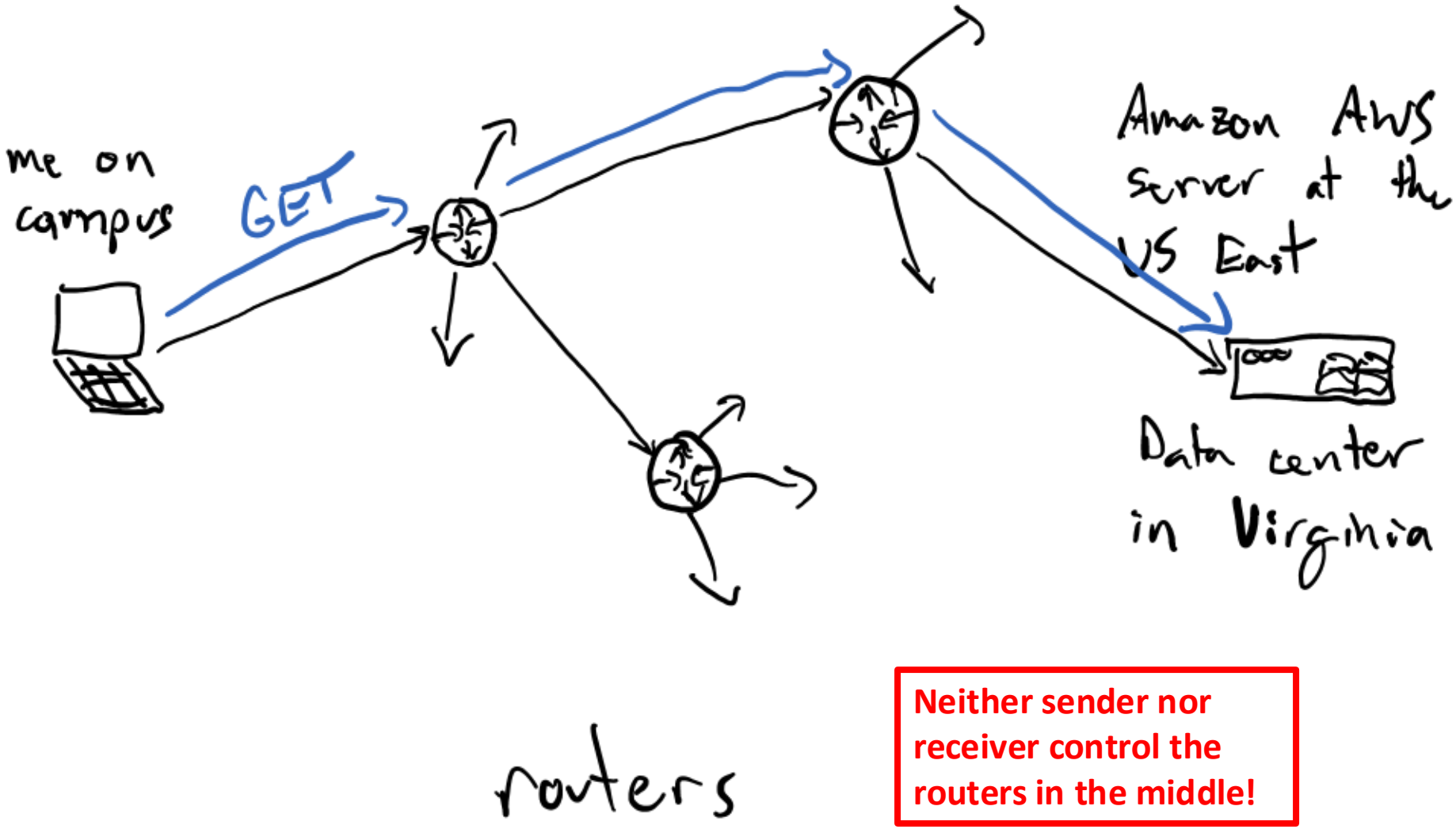


Review: IP Address

- There are lots of computers connected to the internet
- Problem: How to tell them apart?
- Solution: IP Address

- Every computer on the Internet has an address
  - Google 172.217.5.14
  - Amazon 205.251.242.103
  - My laptop 35.7.33.89

Review: Internet routers



Neither sender nor receiver control the routers in the middle!

Reivew: Eavesdropping

- Someone else listens to what you send
- HTTPS means they can't understand it, but...

- NSA data center near SLC: "An article by Forbes estimates the storage capacity as between 3 and 12 exabytes in the near term, based on analysis of unclassified blueprints" (Wikipedia)

The New York Times

N.S.A. Buys Americans' Internet Data Without Warrants, Letter Says

The disclosure comes amid congressional scrutiny and a Federal Trade Commission crackdown on commercial data brokers.

The National Security Agency buys certain logs related to Americans' domestic internet activities from commercial data brokers, according to an unclassified letter by the agency.

January 25, 2024

Review: Network security

- Two parties communicate over a network
- Assume powerful adversary
  - Can read (eavesdrop on) all data transmitted
  - Can modify or delete any data
  - Can inject new data

- Communication: What properties would you like?

Review: Desirable properties

- Confidentiality
  - Adversary should not understand message
  - Solved with symmetric encryption
- Sender authenticity
  - Message is really from the purported sender
  - Solved with asymmetric encryption and public key infrastructure (PKI)
- Message integrity
  - Message not modified between send and receive
  - Solved with symmetric encryption

Review: Desirable properties

- Freshness
  - Message was sent “recently”
  - Solved with symmetric encryption and timestamp, sequence number or nonce
- Anonymity
  - Attacker should not know that we are communicating
  - Solved with VPN (weak) or Tor (strong)

Motivation

- We've already solved the problem of private message content
  - Symmetric and asymmetric encryption
- Today we'll solve the problem of **private message metadata**
  - Source IP and destination IP

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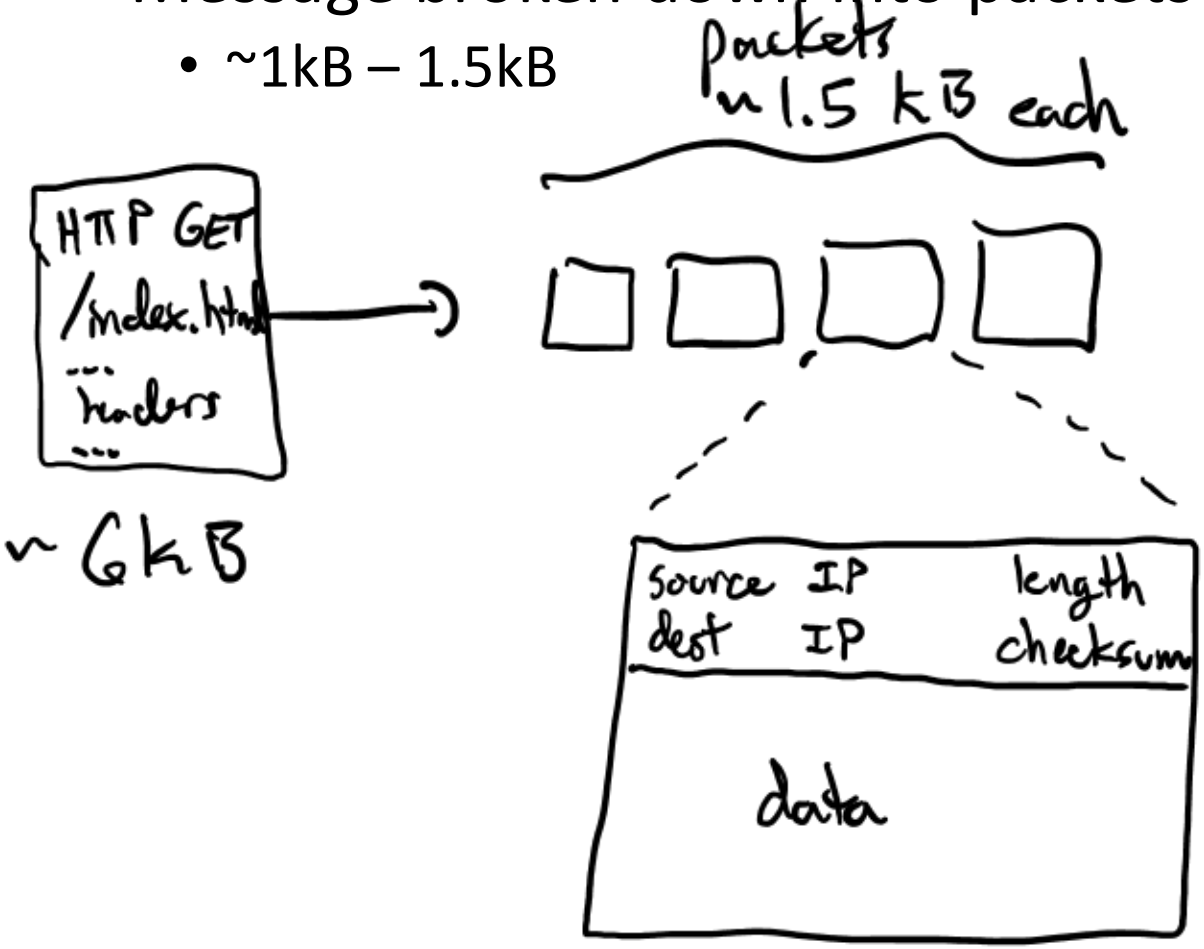
Review: Packets

- Message broken down into packets by sender
  - ~1kB – 1.5kB



Packet inspection

- Each packet has source IP & destination IP
- If you change destination, routers won’t deliver it to the right place
- Can sometimes change source (spoofing) but not always

Packet inspection

- Packet source and destination are visible to anyone observing the network

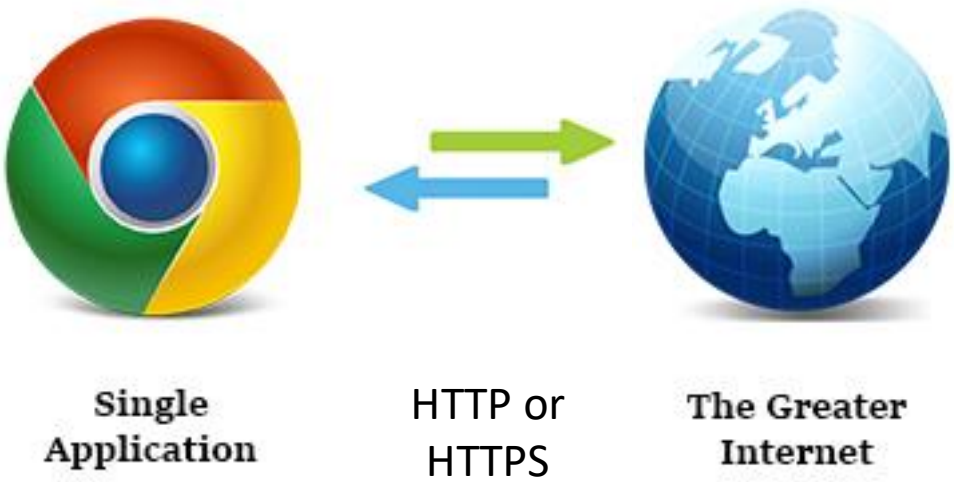


Image: https://www.howtogeek.com

Metadata

- Packet source and destination are visible to anyone observing the network
  - Your ISP
  - Destination web site
  - Any intermediate router
  - Network eavesdropper
- Simply knowing two parties are communicating reveals info
  - Metadata

Metadata

- They know you rang a phone sex line at 2:24 am and spoke for 18 minutes. But they don't know what you talked about.
- They know you called the suicide prevention hotline from the Golden Gate Bridge. But the topic of the call remains a secret.
- They know you got an email from an HIV testing service, then called your doctor, then visited an HIV support group website in the same hour. But they don't know what was in the email or what you talked about on the phone.
- They know you received an email from a digital rights activist group with the subject line “Let’s Tell Congress: Stop SESTA/FOSTA” and then called your elected representative immediately after. But the content of those communications remains safe from government intrusion.
- They know you called a gynecologist, spoke for a half hour, and then called the local abortion clinic’s number later that day.

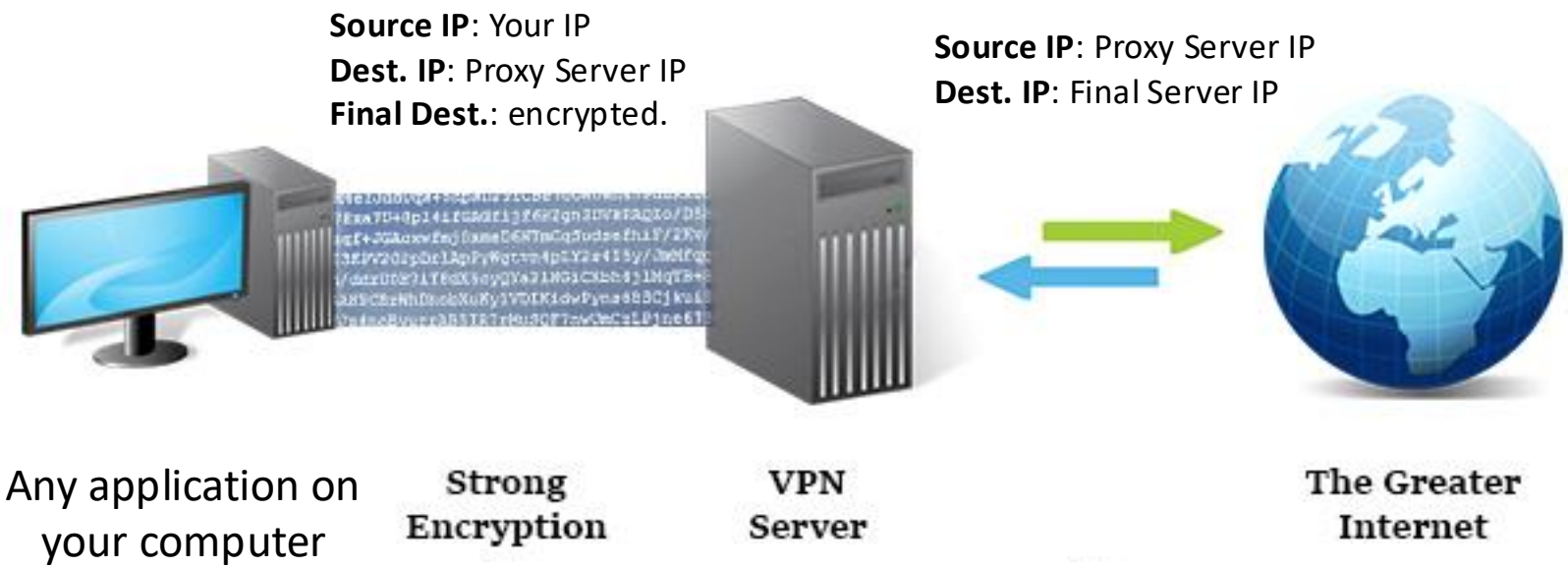
https://ssd.eff.org/en/module/why-metadata-matters

Proxy server

- Proxy server is a middleman
- Makes your internet traffic appear to come from somewhere else
- Hides your IP address

VPN proxy server

- VPN proxy server hides your IP address and encrypts your traffic



VPN proxy server

- Eavesdropper can see client communicate with VPN proxy
- Eavesdropper can see VPN communicate with Web site

VPN proxy server vulnerabilities

- Single point of failure: VPN knows all
- VPN proxy servers vulnerable to subpoenas
  - Court asks proxy server admin for server logs
- Some VPNs have a "destroy logs" policy
  - You'd have to trust your VPN

VPN proxy server vulnerabilities

- VPN proxy servers vulnerable to misconfiguration
- It's easy to screw up VPN server configuration
  - Out-of-date software with security vulnerabilities leads to exfiltrating data
  - Doesn't support IPv6, and IPv6 requests are leaked
  - ... and many others

https://authentic8.blog/vpn-privacy-what-nobody-told-you/

VPNalyzer

- Tool from UM to evaluate VPN security
- Only 4 recommended vendors!
  - Mullvad
  - PIA
  - IVPN
  - Mozilla VPN

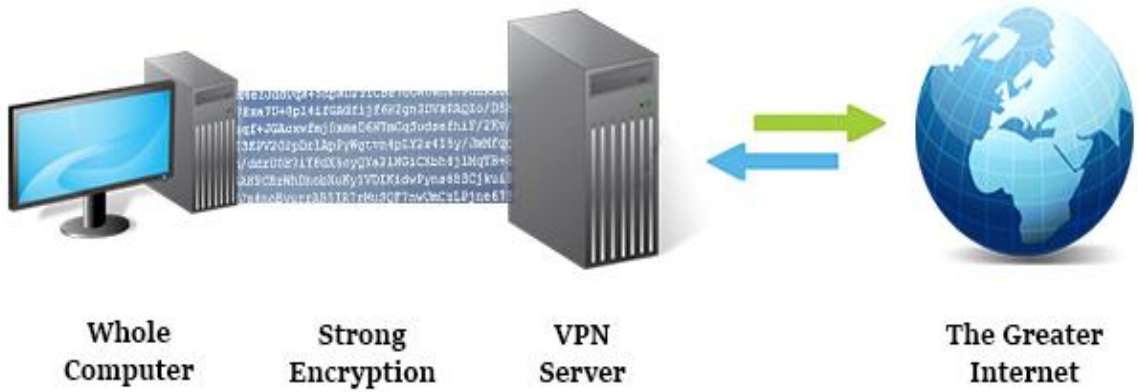
https://vpnalyzer.org/
https://digital-lab-wp.consumerreports.org/wp-content/uploads/2021/12/VPN-White-Paper.pdf

VPN proxy server vulnerabilities

- VPN proxy servers are vulnerable to *traffic analysis*
  - Even if you can trust your VPN and it's configured correctly!
- What if every time client transmits, web site soon gets data?
- Your ISP or other eavesdroppers can use traffic analysis even with a VPN and SSL

Traffic analysis

- Every time you transmit, VPN transmits
  - Gather a bunch of data
- Only need to monitor two points on the network
  - Correlate these two



Tor vs. VPN details: https://2019.www.torproject.org/docs/faq.html.en#IsTorLikeAVPN

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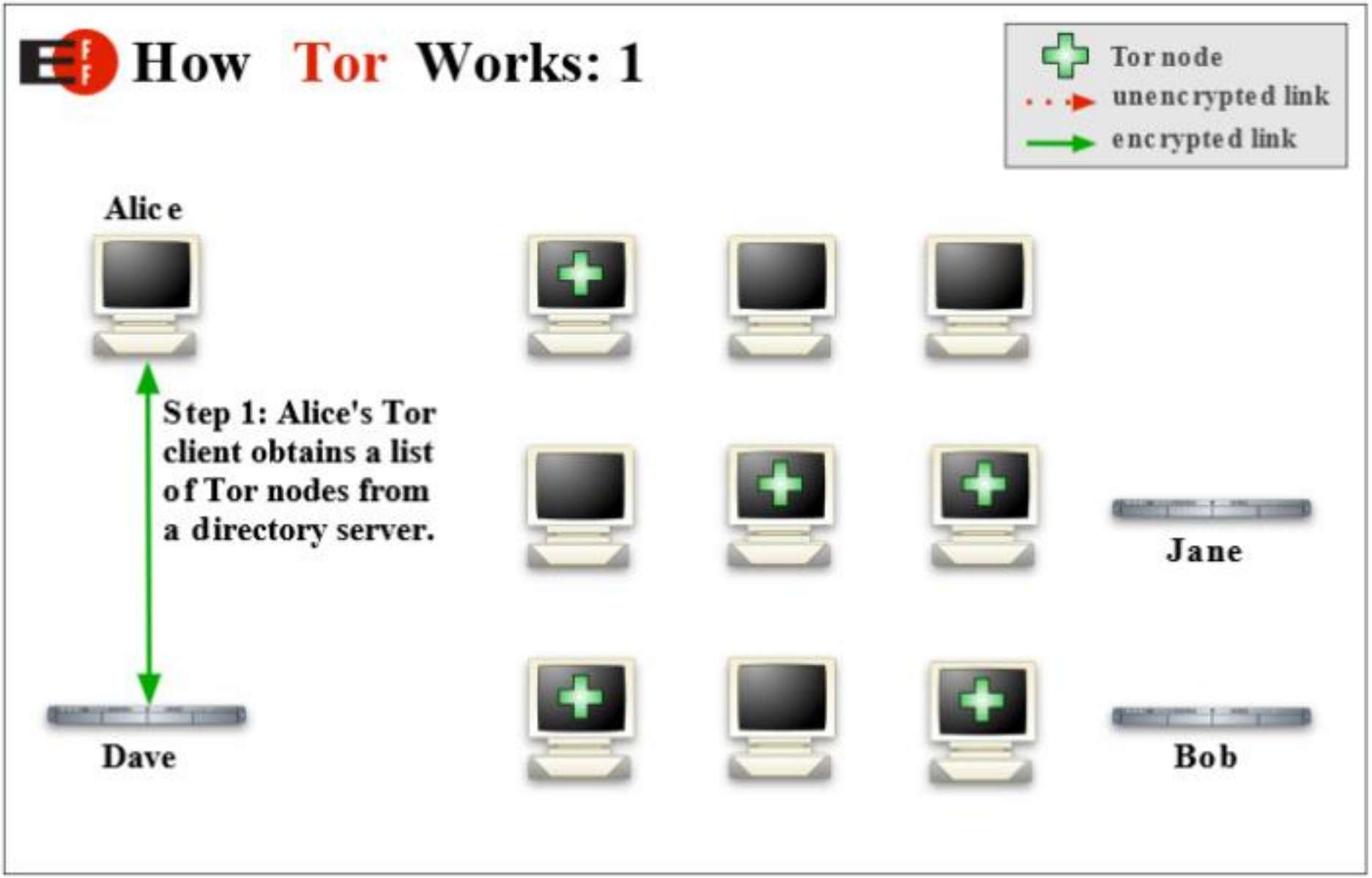
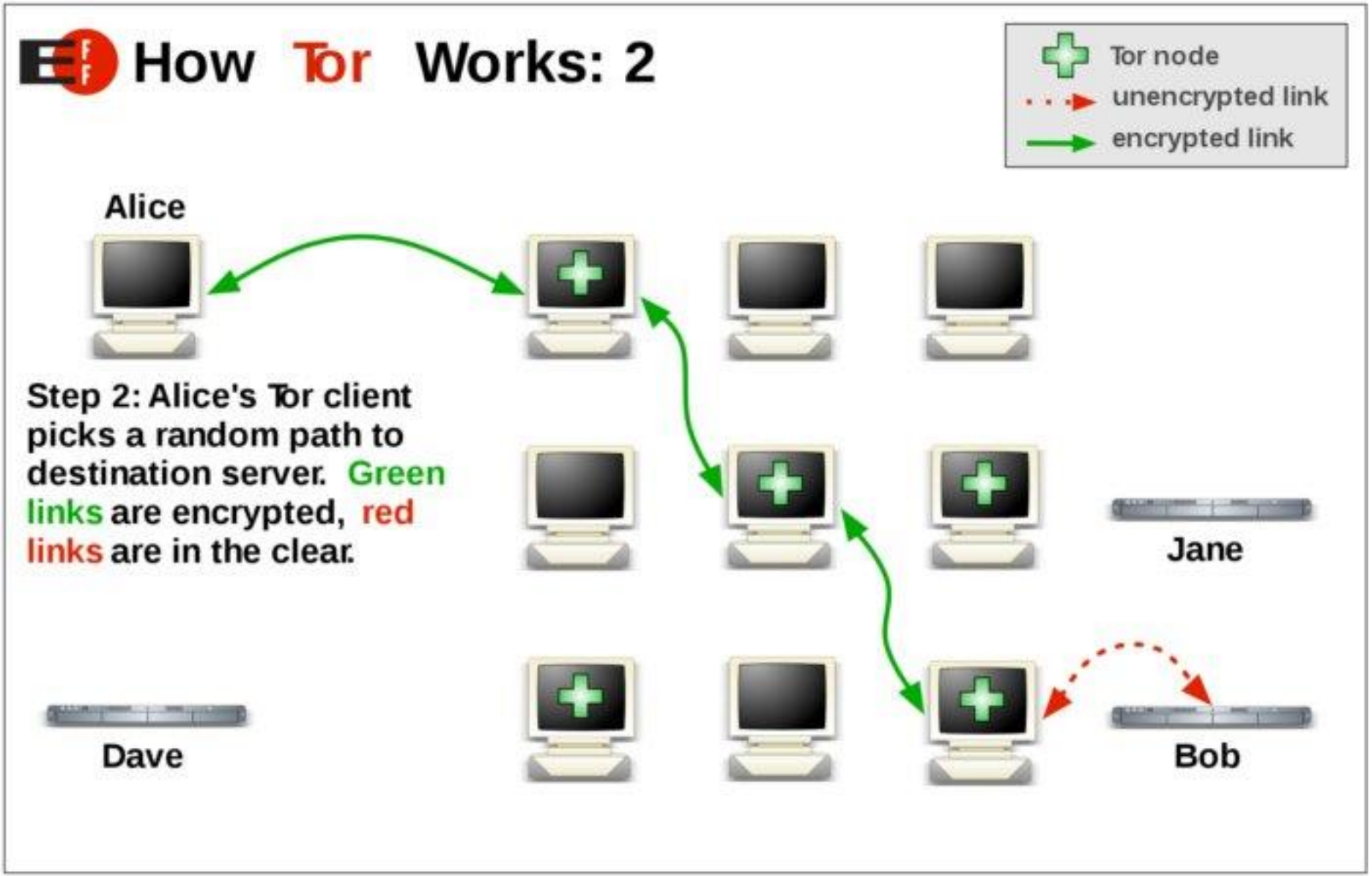
Tor

- **T**he **O**nion **R**outer
- Used to fight traffic analysis



More detail: https://2019.www.torproject.org/about/overview.html.en#thesolution



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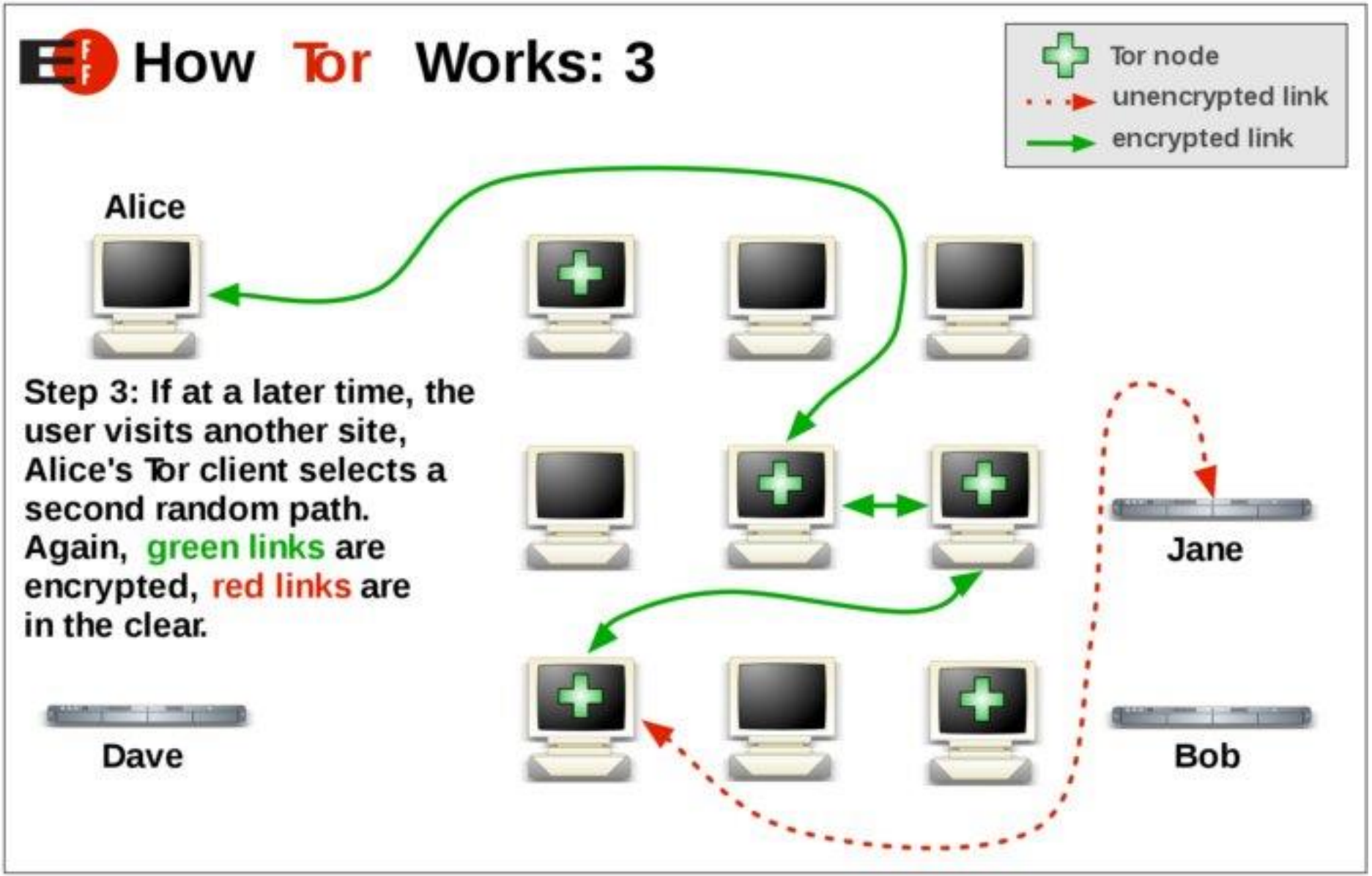

More detail: https://2019.www.torproject.org/about/overview.html.en#thesolution

Tor

- Tor encrypts outgoing data multiple times, like layers of an onion
  - One encryption layer for each step in the relay
- Each layer of encryption is peeled off by a relay node in the network. At each layer:
  - Decrypt the current layer
  - Forward to the next destination

Tor vulnerabilities

- Only the first relay node knows the source IP
- Only the last relay node knows the destination IP
- To break anonymity, you need to surveil many nodes in the Tor circuit

GIZMODO

PRIVACY AND SECURITY

Someone Is Running Hundreds of Malicious Servers on the Tor Network and Might Be De-Anonymizing Users

New research shows a threat actor has taken over a substantial portion of the privacy-protecting service.

https://gizmodo.com/someone-is-running-hundreds-of-malicious-servers-on-the-1848156630

Statistical correlation attack

- Let's say that Elmo and Abby use Tor regularly
  - You control their ISP and collect frequent traffic logs with timestamps
- Also control the ISP for ILoveBigBird.com and DownWithBigBird.org
  - You also collect frequent traffic logs with timestamps
- How can you tell who is pro-BigBird vs anti-BigBird?
  - With enough log data, you can perform statistical correlation attack

Statistical correlation attack

- How to prevent statistical correlation attack?
- Elmo and Abby always transmit data to Tor once per second
- If no data to send, just send NULL or random data
- More Tor users conducting more activity on Tor reduces vulnerability

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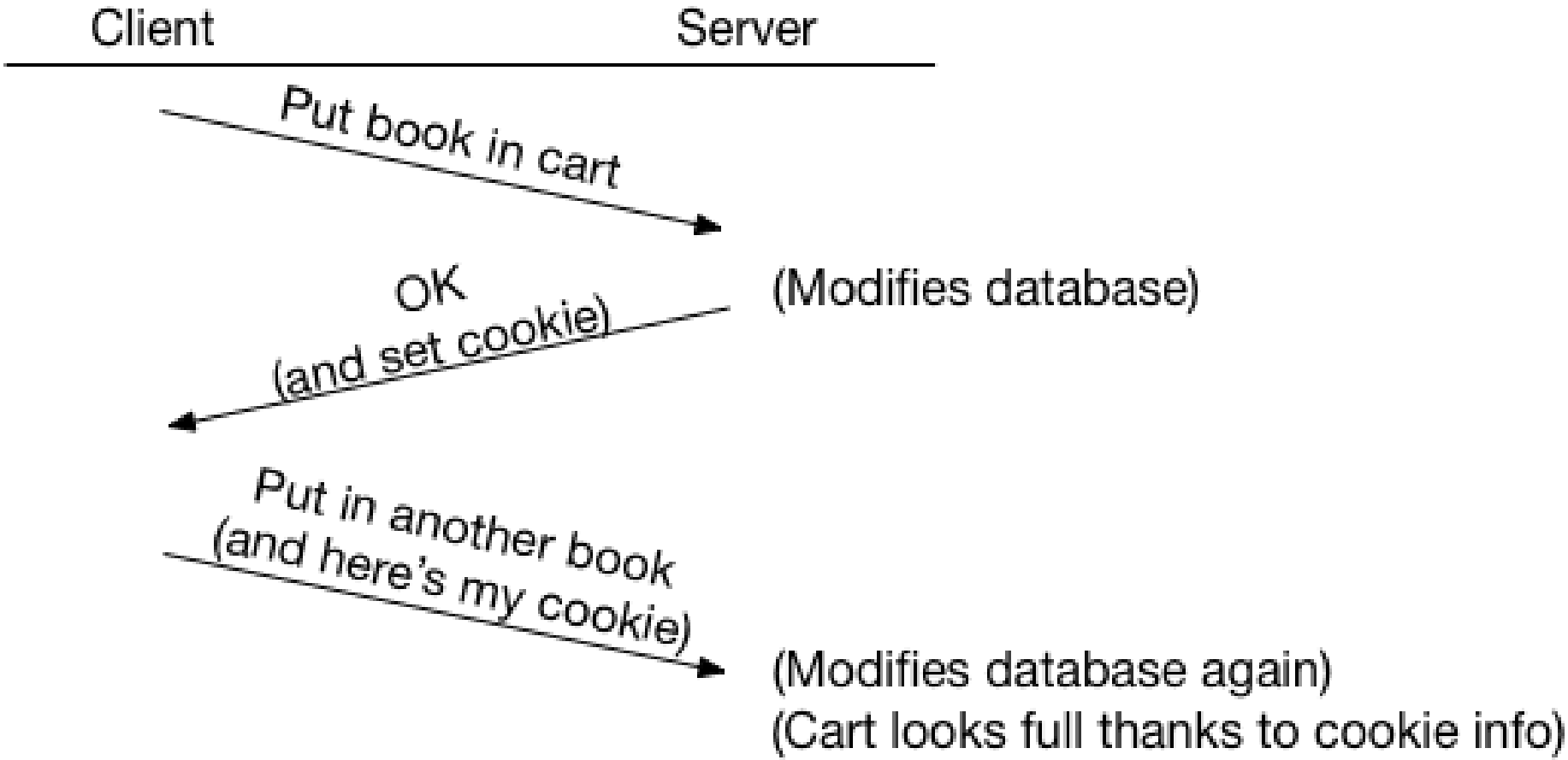
Missing link

- Client can browse anonymously with Tor connection
- Server can publish anonymously with Tor services
- Is that enough?

- Session cookies, including third party trackers are ubiquitous
- Your browser reveal you

Cookies

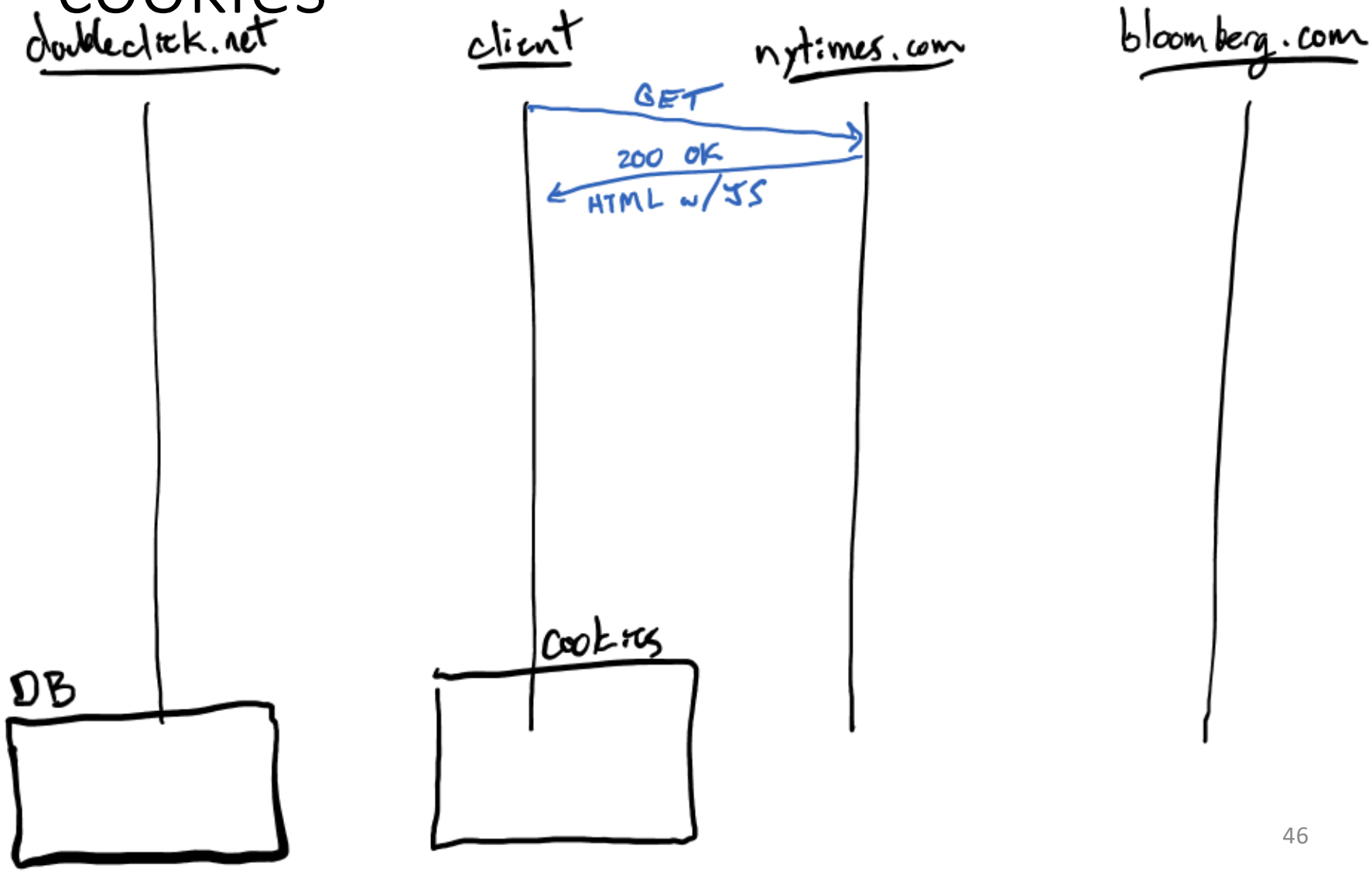

Client
Server
Put book in cart
OK
(and set cookie)
(Modifies database)
Put in another book
(and here's my cookie)
(Modifies database again)
(Cart looks full thanks to cookie info)

Third-party cookies

- Page may contain objects from many sources
  - Scripts, images, etc.
- These 3rd-party objects set and get cookies
- Example: nytimes.com

```
<html>
  <head>
    <script
      src="https://tags.bluekai.com/site/50550?ret=js&limit=1"
      type="text/javascript"
    >
    </script>
  </head>
  <body>
    <img src="https://static01.nyt.com/newsgraphics/2018/01/18/shutdown-deal-
factions/assets/images/lindsey-graham.png">
  </body>
</html>
```

Example: Google third party cookies



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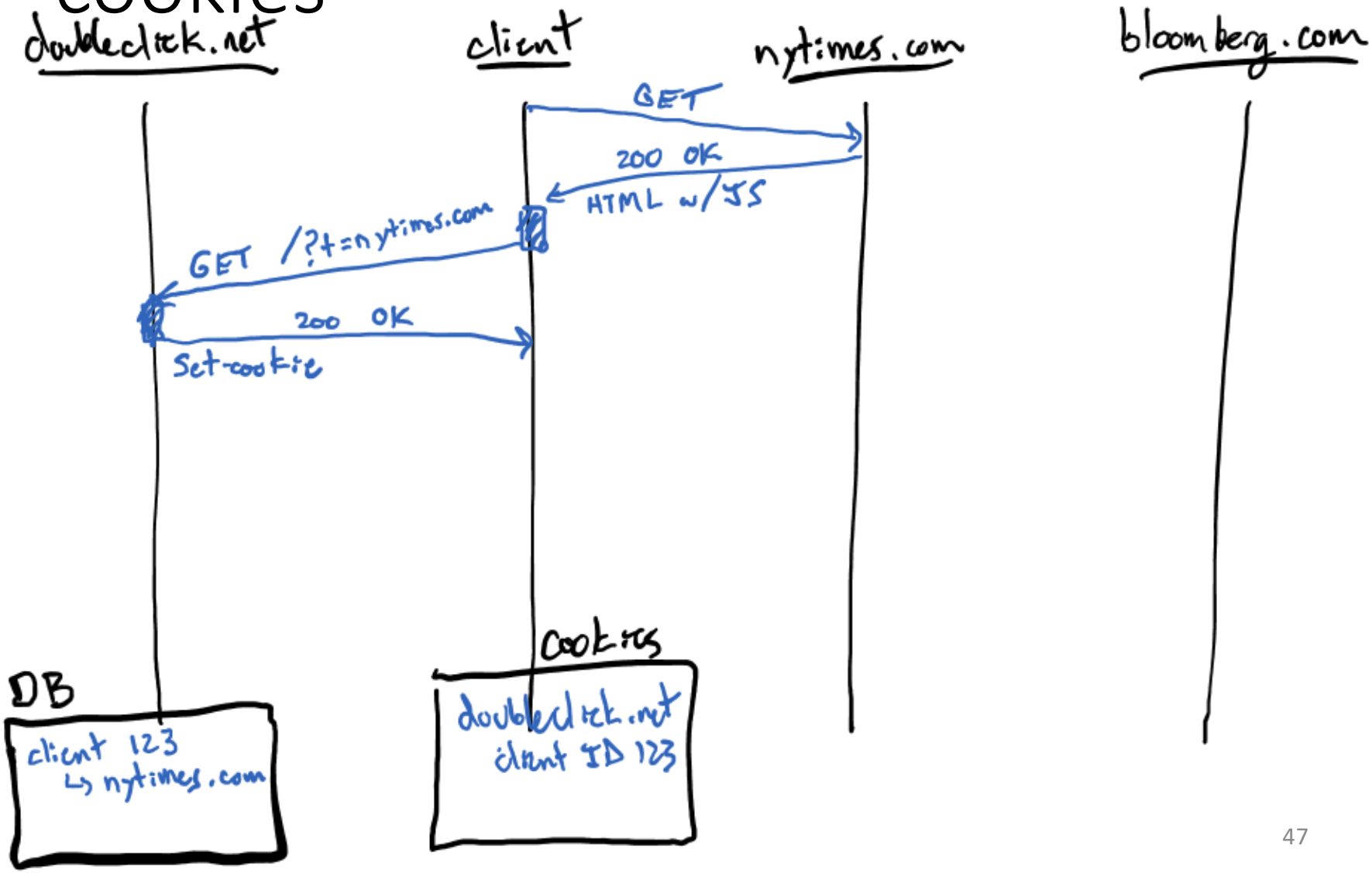

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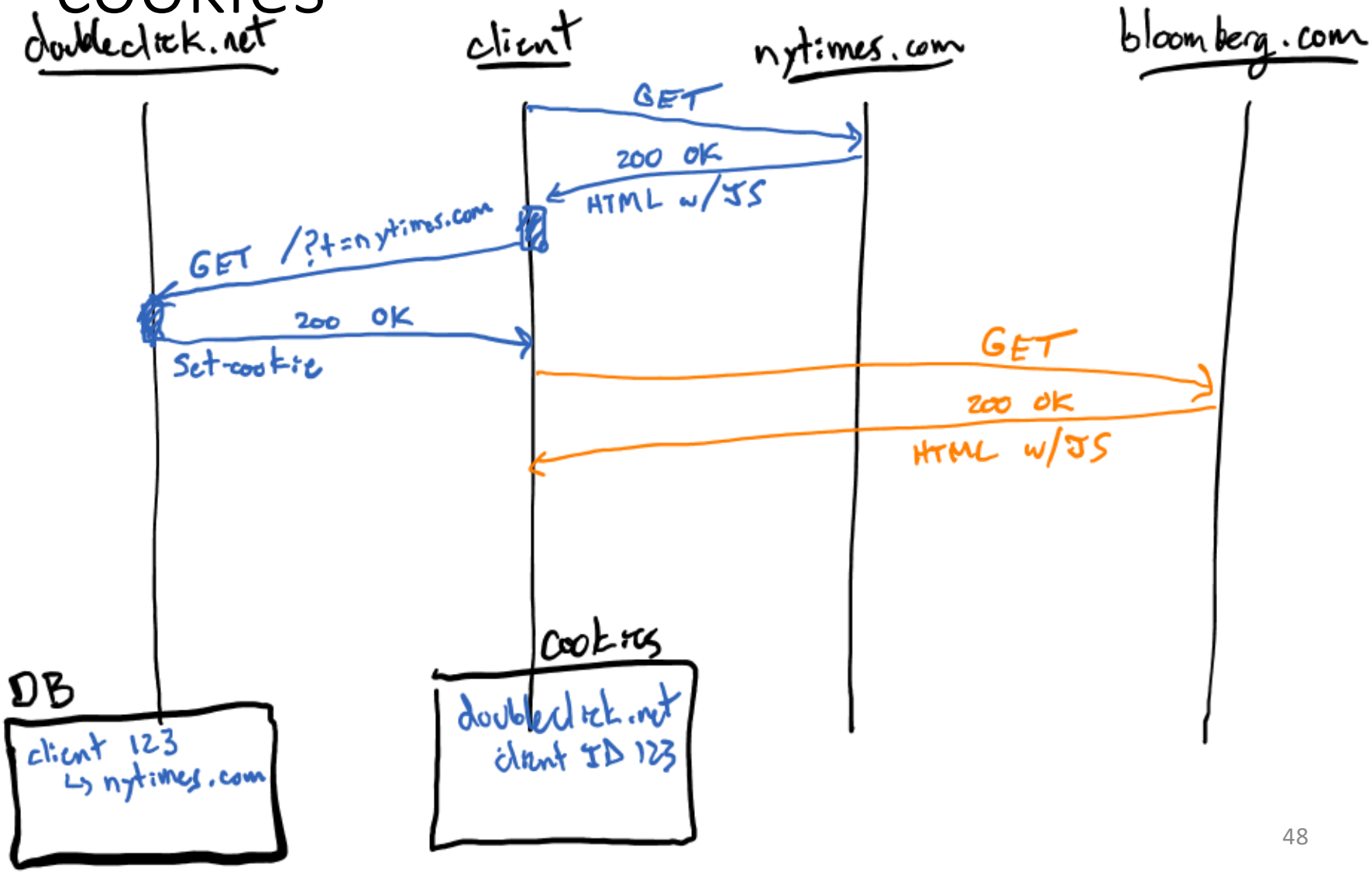

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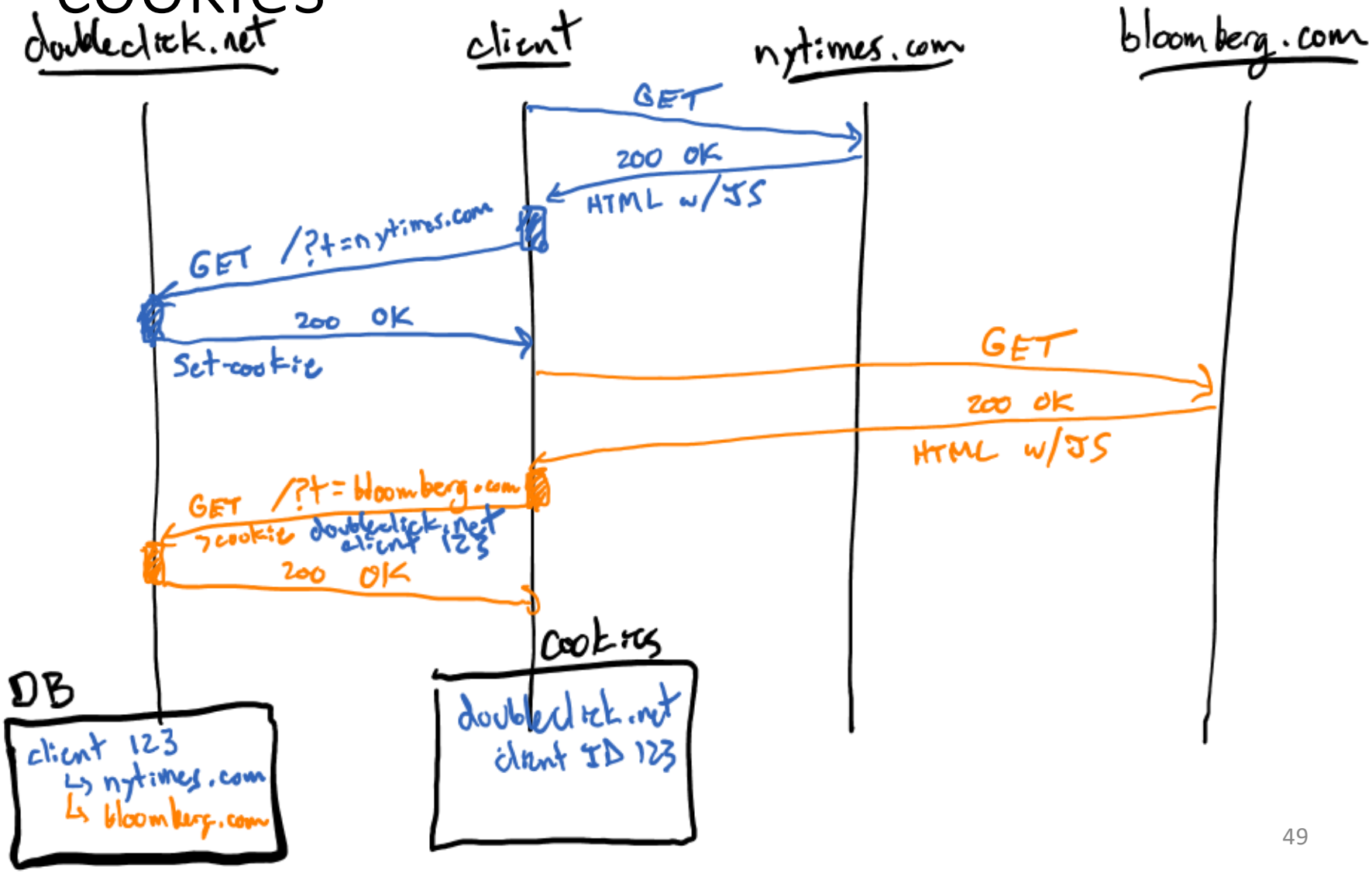
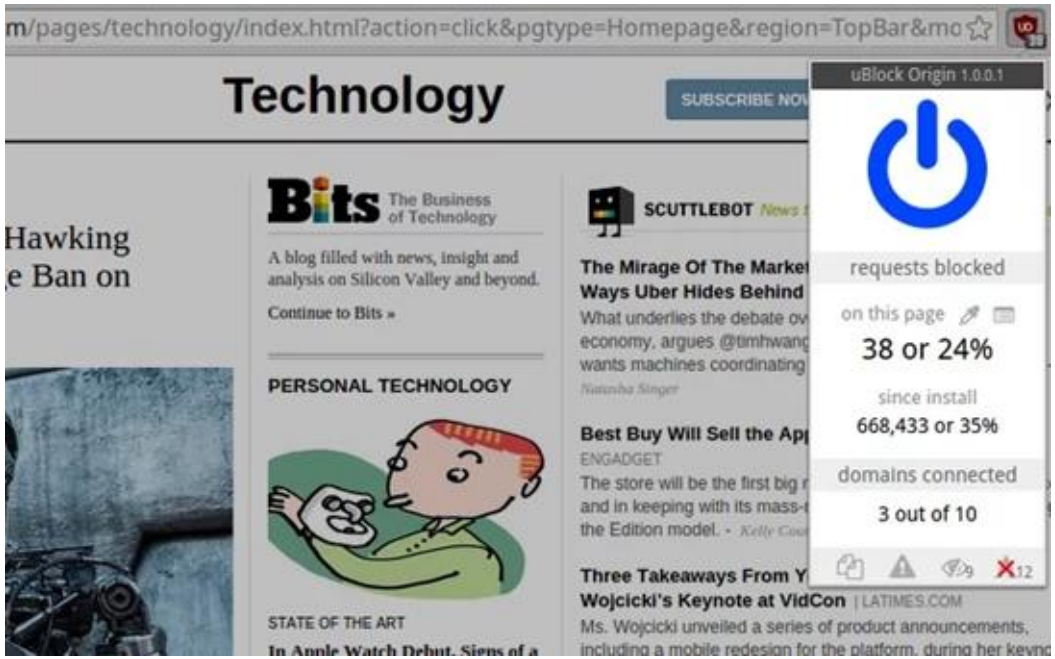


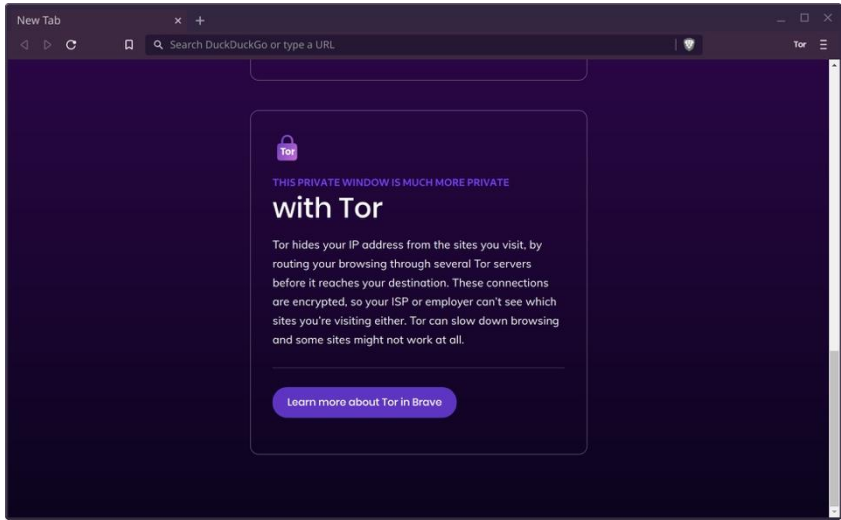
Avoiding trackers

- Add-ons like uBlock Origin, Privacy Badger, Brave, Disconnect or ScriptNo block trackers
  - Monitors embedded links and blacklists trackers



Tor browser

- Routes all traffic via Tor
  - Modified Firefox
- Essential privacy plugins
  - NoScript and HTTPS Everywhere
- Non-tracking search engine
  - DuckDuckGo
- Alternative: Brave Browser ships with Tor integration



https://www.torproject.org/download/

Browser fingerprinting

- Torbrowser avoids browser fingerprinting
- Browser fingerprinting: collect enough information from a client, you can (probably) identify them uniquely
  - User-Agent string
  - Screen size
  - Browser plugin details
  - Language
  - Platform
  - Etc.

https://www.torproject.org/download/

Not anonymous browsing

- We've talked about *more* anonymous browsing
  - Tor Browser and Duck Duck Go are a good start
- What about *less* anonymous browsing?
  - Google Chrome tracks you, even in incognito mode
  - Google search engine tracks you with Google login, third party cookies *and* Chrome!

- Middle ground:
  - Brave Browser or Firefox
  - Duck Duck Go instead of Google

Browser recommendations

- Brave
- Firefox
- Safari

- Duck Duck Go search engine

- Better privacy protections, even without Tor

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Anonymous servers

- So far, we've talked about a client can browse anonymously

- Problem: How do you run a server anonymously?
- Solution: Tor services

Tor services

- Tor allows users to anonymously publish services
  - E.g., web pages, or a chat server
- Service locations (*rendezvous points*) must be known to clients
  - Even though censors may want to locate, take down services
- Key idea: layer of indirection
  - *Introduction points* are Tor nodes that relay traffic from clients to services

More detail: https://2019.www.torproject.org/docs/onion-services.html

Finding Tor services

- Curated lists
  - Reddit
  - Hidden Wiki
  - ... and many others
- Hidden service search engines
  - Ahmia
  - Torch
  - Not Evil
  - ... and many others
- Search engine crawlers on dark web
  - Conventional crawlers can't reach dark web
  - Route crawler GET requests to .onion sites through Tor

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Tor history

- Developed by US Naval Research Laboratory
- Goal: protect US intelligence communications online
- Now a nonprofit
  - US Government is a major funder

More uses for Tor

- Break censorship barriers
  - E.g., in places that censor web content
- Political activism
  - E.g., in places that discourage free speech
- Net neutrality
  - E.g., ISP speeds up some traffic, slows down other traffic
- Reporting abuse or corruption
  - E.g., whistleblower concerned for safety, Tor enables anonymous communication

Summary

- Tor enables anonymity online
- Not perfect, but vastly more anonymous than traditional browsing